

Detection of changes in spatial data

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Change-point detection in spatial data

Plan

- Two examples
- Markov random field
- Explicit likelihood function
- Pseudolikelihood methods
- Pseudo-CUSUM and its performance

Application 1 - Nanotechnology

10 μm	—	1971
6 μm	—	1974
3 μm	—	1977
1.5 μm	—	1982
1 μm	—	1985
800 nm	—	1989
600 nm	—	1994
350 nm	—	1995
250 nm	—	1997
180 nm	—	1999
130 nm	—	2001
90 nm	—	2004
65 nm	—	2006
45 nm	—	2008
32 nm	—	2010
22 nm	—	2012
14 nm	—	2014
10 nm	—	2017
7 nm	—	2018



Application 1 - Nanotechnology

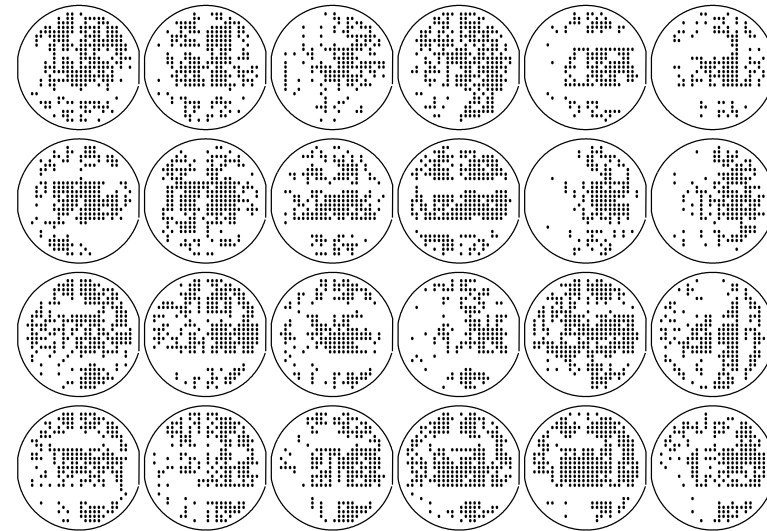
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65 nm	—	2006
45 nm	—	2008
32 nm	—	2010
22 nm	—	2012
14 nm	—	2014
10 nm	—	2017
7 nm	—	2018
5 nm	—	$\approx$ 2020



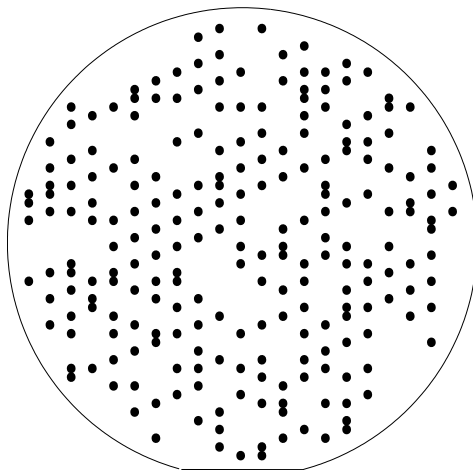
Application 1 - Nanotechnology

Defective chips form clusters of various sizes, shapes, locations.

A change in any spatial parameters indicates a new root cause or a new impact of a root cause.

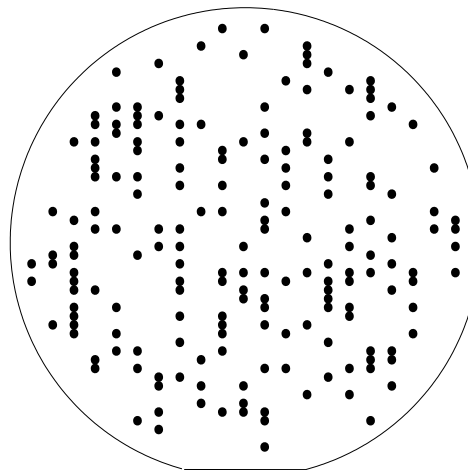


Diagonal patterns generated by Gibbs sampler

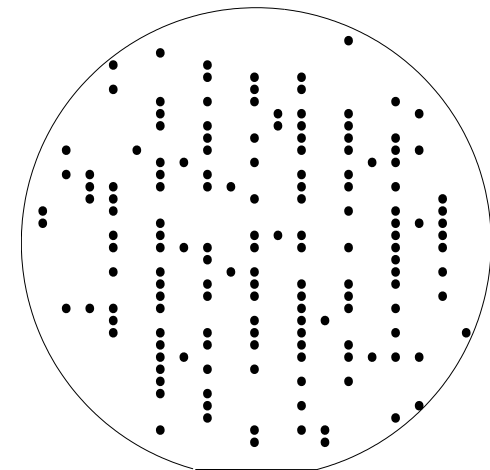


Diagonal patterns

Gibbs sampler with parameters from lot 9935....



Different levels of vertical interaction



Application 2 - Violence, Unrest, Political Events



Baltimore riots: A timeline - CNN
cnn.com



Baltimore riots: A timeline - CNN
cnn.com



Baltimore riots: State of emergency ...
washingtonpost.com



Freddie Gray protests: 5 Fast Facts ...
latimes.com



The Baltimore Riots: Does Violence Have ...
businessinsider.com



declared in Baltimore after riots erupt ...
fortune.com



Media Covered the Baltimore Riots ...
ap.org



Baltimore erupts into violence, chaos ...
nbc12.com

Protests in Baltimore (2015)



Baltimore riots: A timeline - CNN
cnn.com



Baltimore riots: A timeline - CNN
cnn.com



Baltimore riots: State of emergency -
washingtonpost.com



Freddie Gray Protests: 5 Fast Facts -
baltimoresun.com



The Baltimore Riots: Does Violence Have ...
cnn.com



declared in Baltimore after riots erupt ...
fortune.com



Media Covered the Baltimore Riots -
ap.org



Baltimore (Riots into Violence, Chaos ...
fox26.com

Civil protests in Baltimore on April 18 - May 3, 2015, following the death of Freddie Gray. State of emergency, curfew, twenty police officers injured, at least 250 people arrested, approximately 300 damaged businesses, 150 vehicle fires, 60 house fires, 27 looted stores looted, \$9 mln. in damages.



protest over ex-cop's acquittal ...
abcnews.go.com



Jason Stockley: St. Louis ex-officer ...
cnn.com



St. Louis protests: More than 80 ...
cnn.com



front lines in St. Louis ...
abcnews.go.com



protest over ex-cop's acquittal ...
abcnews.go.com



After Jason Stockley not-guilty verdict ...
cbsnews.com



ex-St. Louis officer's acquittal in killing ...
mercurynews.com



St. Louis after ex-cop acquitted ...
twincities.com

Unrest in St. Louis (2017)



protest over ex-cop's acquittal ...
abcnews.go.com



Jason Stockley: St. Louis ex-officer ...
cnn.com



St. Louis protests: More than 80 ...
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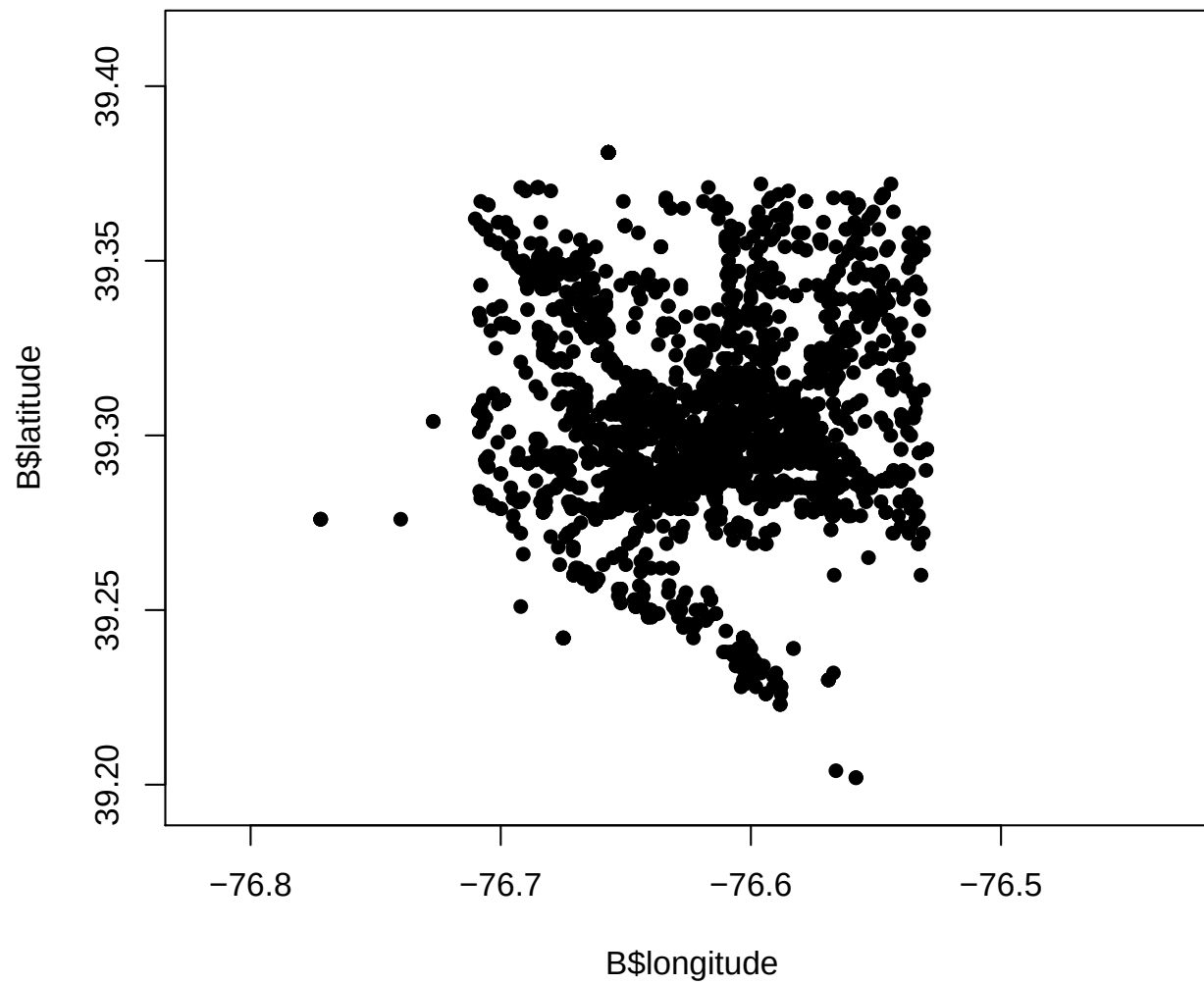


ex-St. Louis officer's acquittal in killing ...
mercurynews.com

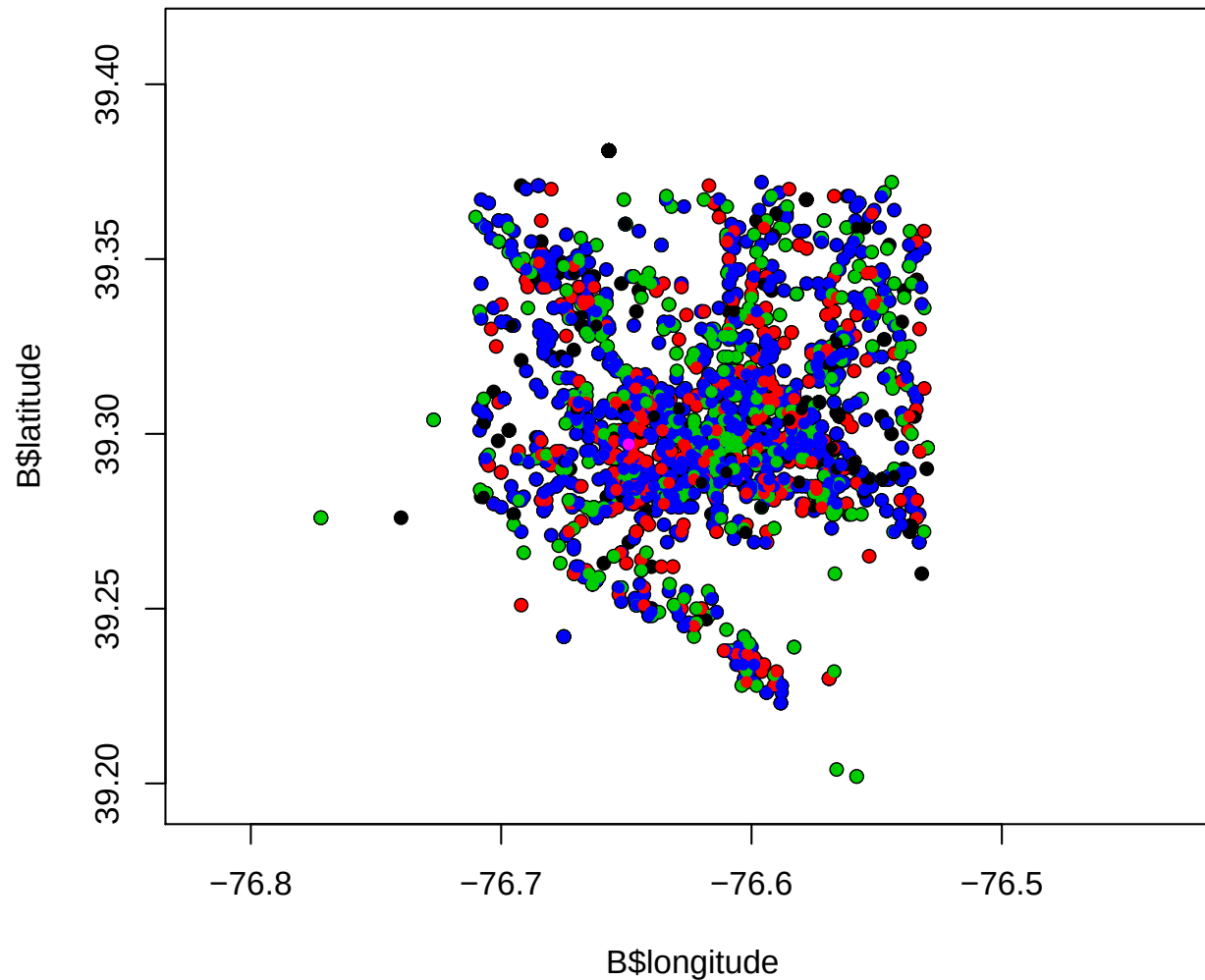


St. Louis after ex-cop acquitted ...
twincities.com

Protests in St. Louis and Ferguson in September–November 2017, after a former police officer was found not guilty in the 2011 shooting of Anthony Lamar Smith. Eleven police officers injured, 33 people arrested.



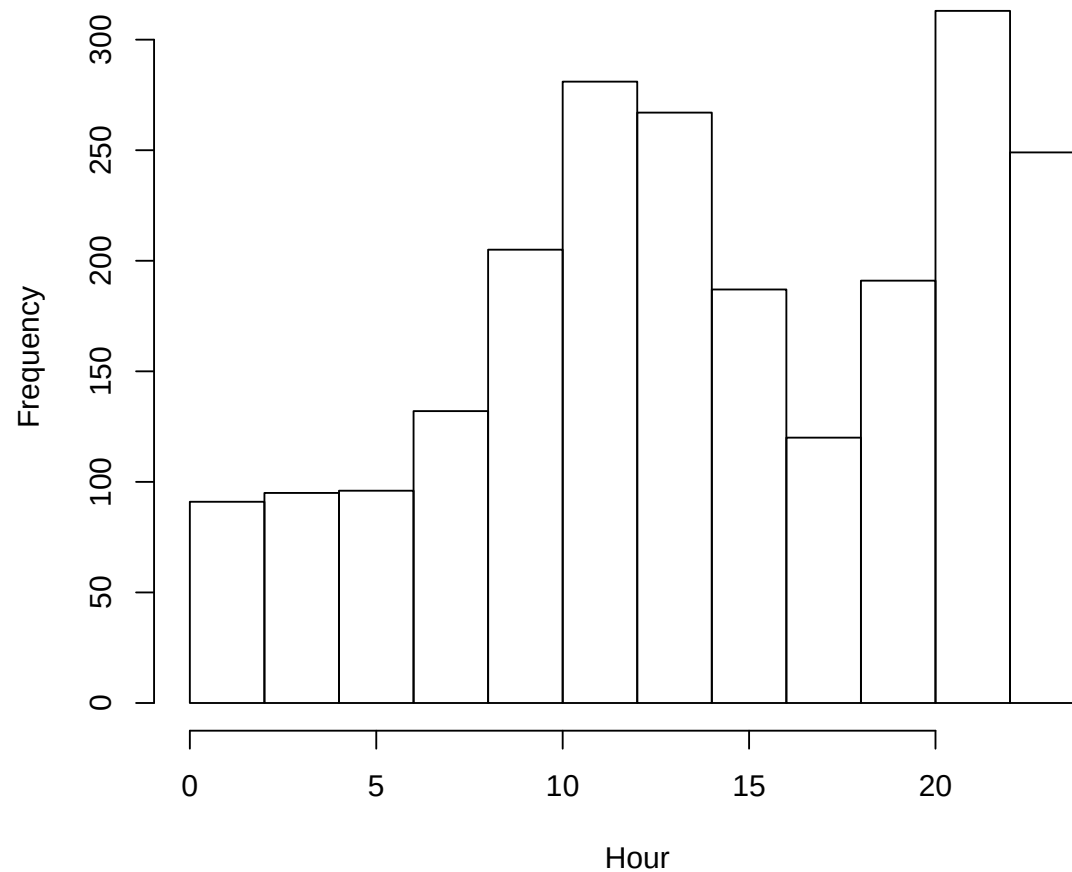
911 calls in Baltimore, 08/18/2017



Priority	Non-emergency	Low	Medium	High	Emergency	Total
N	329	614	1382	515	2	2842

911 calls in Baltimore, 08/18/2017

Daily dynamics



Vegan Restaurant Week. Los Angeles Angels @ Baltimore Orioles (7:9).

Spatial modeling. Markov random field

Markov chain: the future depends on the past through the present

$$P \{X_{t+h} \mid X_s, s \leq t\} = P \{X_{t+h} \mid X_t\}$$

Markov random field: each location depends on the rest of the surface through the neighborhood Q

$$P \{X_{ij} \mid \mathcal{W} \setminus (i, j)\} = P \{X_{ij} \mid Q_{ij}\}$$

$$\mathcal{W} = \{X_{ij}\} = \text{surface}$$

$$X_{ij} = \begin{cases} 1 & \text{if the event happens at location } (i, j) \\ 0 & \text{otherwise} \end{cases}$$

$$Q_{i,j} = \text{neighborhood of the location } (i, j)$$

Four-unit neighborhood

Neighborhood: $\mathcal{Q}_{ij} = \{(k, l) : |k - i| + |l - j| = 1\}.$

		$t(s)$	
	$l(s)$	s	$r(s)$
		$b(s)$	

Leads to an *auto-logistic model*

Besag (1974)

- ▶ simple, tractable
- ▶ does not model diagonal patterns

Eight-unit neighborhood

Neighborhood: $\mathcal{Q}_{ij} = \{(k, l) \neq (i, j) : |k - i| \leq 1, |l - j| \leq 1\}.$

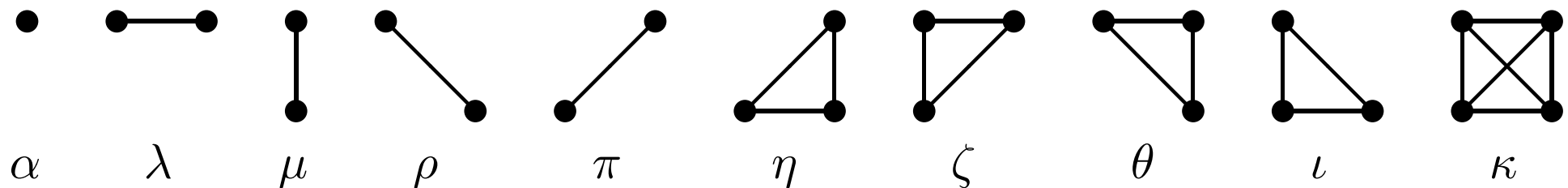
	$lt(s)$	$t(s)$	$rt(s)$
	$l(s)$	s	$r(s)$
	$lb(s)$	$b(s)$	$rb(s)$

- ▶ Joint distribution of $\{X_{ij}\} = ???$
- ▶ Conditional distribution of $X_{ij} \mid \mathcal{Q}_{ij} = ?$
- ▶ Parameter estimation and explanation of spatial patterns
- ▶ Change-point detection
- ▶ Connection to our applications

Parameters of a Markov random field

Clique = a set of chips in which every chip is a neighbor of every other chip

There are ten types of cliques –



Let

$$\mathcal{Q}'_s \subset \mathcal{Q}_s,$$

$$L_s(\mathcal{Q}'_s) = \text{logit} \left(P \left\{ X_s = 1 \mid \begin{array}{l} X_t = 1 \text{ for } t \in \mathcal{Q}'_s \\ X_t = 0 \text{ for } t \in \mathcal{Q}_s \setminus \mathcal{Q}'_s \end{array} \right\} \right).$$

10 cliques $\implies$ 10 parameters of MRF

$$\alpha = L(\emptyset)$$



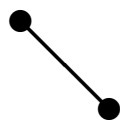
α



λ



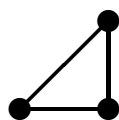
μ



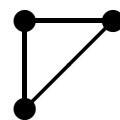
ρ



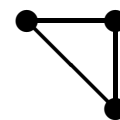
π



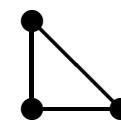
η



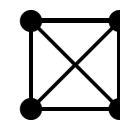
ζ



θ



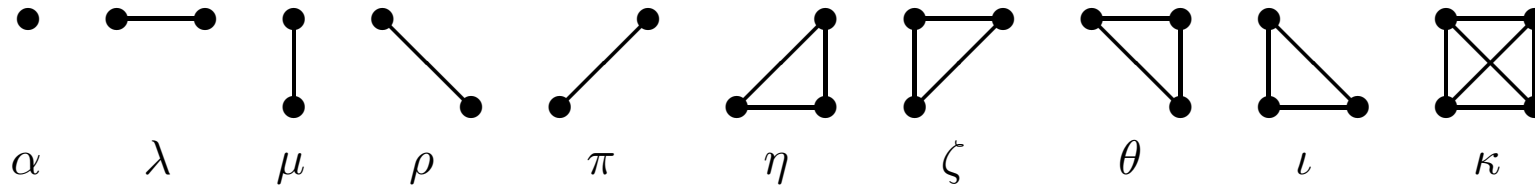
ι



κ

10 cliques $\implies$ 10 parameters of MRF

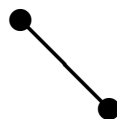
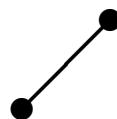
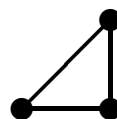
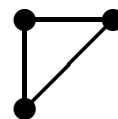
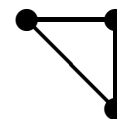
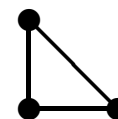
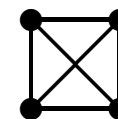
$\alpha = L(\emptyset) =$ logit of a conditional probability to fail
given that all surrounding chips are good



10 cliques $\implies$ 10 parameters of MRF

$$\alpha = L(\emptyset)$$

$\lambda = L(r) - \alpha = L(l) - \alpha =$ *additional logit due to the failure of one chip on the left or on the right*

 α  λ  μ  ρ  π  η  ζ  θ  ι  κ

10 cliques $\implies$ 10 parameters of MRF

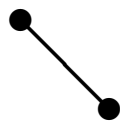
$$\alpha = L(\emptyset)$$

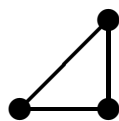
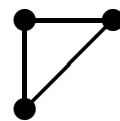
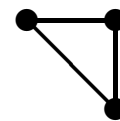
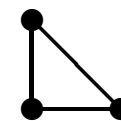
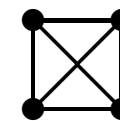
$$\lambda = L(r) - \alpha = L(l) - \alpha$$

$$\mu = L(t) - \alpha = L(b) - \alpha = \text{additional logits due to the failure of one chip below or above}$$


 α

 λ

 μ

 ρ

 π

 η

 ζ

 θ

 ι

 κ

10 cliques $\implies$ 10 parameters of MRF

$$\left\{ \begin{array}{l} \alpha = L(\emptyset) \\ \lambda = L(r) - \alpha = L(l) - \alpha \\ \mu = L(t) - \alpha = L(b) - \alpha \\ \begin{array}{l} \rho = L(rb) - \alpha = L(lt) - \alpha \\ \pi = L(rt) - \alpha = L(lb) - \alpha \end{array} \end{array} \right\} \begin{array}{l} \\ \\ \\ \text{additional logits due to a failed} \\ \text{diagonal neighbor} \end{array}$$

$$\left\{ \begin{array}{ccccccccc} \bullet & \bullet\text{---}\bullet & \bullet & \bullet & \bullet & \triangle & \triangle & \triangle & \triangle & \square \\ \alpha & \lambda & \mu & \rho & \pi & \eta & \zeta & \theta & \iota & \kappa \end{array} \right.$$

10 cliques $\implies$ 10 parameters of MRF

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$$\theta = L(l, b) - \lambda - \mu - \alpha = L(r, rb) - \lambda - \rho - \alpha = L(t, lt) - \mu - \rho - \alpha$$

$$\iota = L(r, t) - \lambda - \mu - \alpha = L(l, lt) - \lambda - \rho - \alpha = L(b, rb) - \mu - \rho - \alpha$$

= additional logits due to the *simultaneous* failure of two neighbors



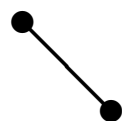
α



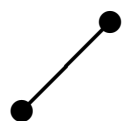
λ



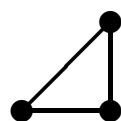
μ



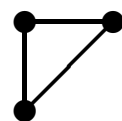
ρ



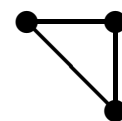
π



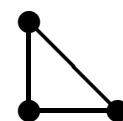
η



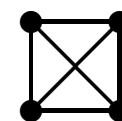
ζ



θ



ι



κ

10 cliques $\implies$ 10 parameters of MRF

$$\alpha = L(\emptyset)$$

$$\lambda = L(r) - \alpha = L(l) - \alpha$$

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$$\iota = L(r, t) - \lambda - \mu - \alpha = L(l, lt) - \lambda - \rho - \alpha = L(b, rb) - \mu - \rho - \alpha$$

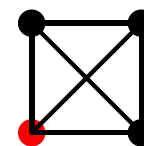
$$\kappa = L(r, t, rt) - \eta - \zeta - \iota - \lambda - \mu - \pi - \alpha$$

$$= L(l, t, lt) - \eta - \theta - \iota - \lambda - \mu - \rho - \alpha$$

$$= L(r, b, rb) - \theta - \zeta - \iota - \lambda - \mu - \rho - \alpha$$

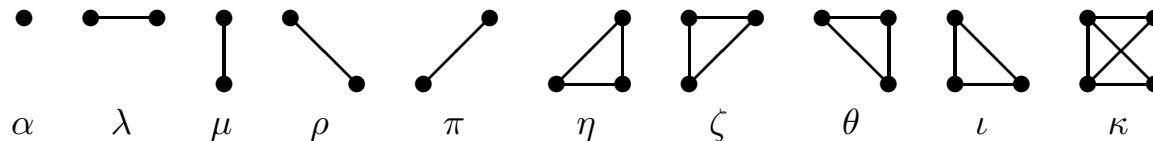
$$= L(l, b, lb) - \eta - \zeta - \theta - \lambda - \mu - \pi - \alpha$$

= additional logits due to the simultaneous failure of *three* neighbors



κ

10 cliques $\implies$ 10 parameters of MRF



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$$\mu = L(t) - \alpha = L(b) - \alpha$$

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$$\iota = L(r, t) - \lambda - \mu - \alpha = L(l, lt) - \lambda - \rho - \alpha = L(b, rb) - \mu - \rho - \alpha$$

$$\kappa = L(r, t, rt) - \eta - \zeta - \iota - \lambda - \mu - \pi - \alpha$$

$$= L(l, t, lt) - \eta - \theta - \iota - \lambda - \mu - \rho - \alpha$$

$$= L(r, b, rb) - \theta - \zeta - \iota - \lambda - \mu - \rho - \alpha$$

$$= L(l, b, lb) - \eta - \zeta - \theta - \lambda - \mu - \pi - \alpha$$

10 cliques $\implies$ 10 parameters $\implies$ 10 statistics

{	$N_\alpha = \sum X_{ij}$										
		α	λ	μ	ρ	π	η	ζ	θ	ι	κ
	$N_\lambda = \sum X_{ij} X_{i+1j}$										
	$N_\mu = \sum X_{ij} X_{ij+1}$										
	$N_\rho = \sum X_{ij} X_{i+1j-1}$										
	$N_\pi = \sum X_{ij} X_{i+1j+1}$										
	$N_\eta = \sum X_{ij} X_{ij+1} X_{i-1j}$										
	$N_\zeta = \sum X_{ij} X_{ij-1} X_{i+1j}$										
	$N_\theta = \sum X_{ij} X_{ij-1} X_{i-1j}$										
	$N_\iota = \sum X_{ij} X_{ij+1} X_{i+1j}$										
$N_\kappa = \sum X_{ij} X_{ij+1} X_{i+1j} X_{i+1j+1}$											

Likelihood function of MRF

By the Hammersley–Clifford Theorem,

$$\log \frac{P(\mathbf{X})}{P(\mathbf{0})} = N_\alpha G_\alpha + N_\lambda G_\lambda + \cdots + N_\kappa G_\kappa \text{ for some functions } G$$

Find $G_{(\cdot)}$ from the Factorization Theorem,

$$\frac{P(\mathbf{X})}{P(\mathbf{Y})} = \prod_{i=1}^n \frac{P\{X_i \mid X_j, j < i; Y_k, k > i\}}{P\{Y_i \mid X_j, j < i; Y_k, k > i\}}$$

Hammersley and Clifford (1971), unpublished

Likelihood function of MRF

Finding G_α .

Let $\begin{cases} \mathbf{Y} = \mathbf{0} \\ \mathbf{X} = \text{a wafer with 1 defective chip } s \end{cases}$

Then

$$\log \frac{P(\mathbf{X})}{P(\mathbf{0})} = G_\alpha = \log \frac{P\{1 \mid X_{s'} = 0, s' \neq s\}}{P\{0 \mid X_{s'} = 0, s' \neq s\}} = \alpha.$$

Likelihood function of MRF

Next, let $\begin{cases} \mathbf{Y} = \mathbf{0} \\ \mathbf{X} \text{ has 2 defective chips } s, r(s) \end{cases}$
then

$$\begin{aligned} \log \frac{P(\mathbf{X})}{P(\mathbf{0})} &= 2G_\alpha + G_\lambda \\ &= \log \frac{P\{1 \mid X_{r(s)} = 1, X_{s'} = 0, s' \neq s, s' \neq r(s)\}}{P\{0 \mid X_{r(s)} = 1, X_{s'} = 0, s' \neq s, s' \neq r(s)\}} \\ &\quad + \log \frac{P\{1 \mid X_{s'} = 0, s' \neq s\}}{P\{0 \mid X_{s'} = 0, s' \neq s\}} \\ &= 2\alpha + \lambda \end{aligned}$$

Hence $G_\lambda = \lambda$, etc.

Likelihood function of MRF

- The joint distribution is $P(\mathbf{X}) = P(\mathbf{0}) \exp \{ \alpha N_\alpha + \lambda N_\lambda + \dots + \kappa N_\kappa \}$
where $P(\mathbf{0}) = \left(\sum_{\text{surfaces } \mathcal{W}} \exp \{ \alpha N_\alpha + \lambda N_\lambda + \dots + \kappa N_\kappa \} \right)^{-1}$

The normalizing constant is a sum over all $2^{|\mathcal{W}|}$ combinations of 0s and 1s on a surface $\mathcal{W}$. No closed form.

- Conditional distributions (at each location, given its neighborhood) are

$$P \{ X_s | \mathcal{Q}_s \} = \frac{1}{1 + \exp \{ -\alpha \Delta_\alpha(s, X_s) - \dots - \kappa \Delta_\kappa(s, X_s) \}}$$

where $\Delta_{\mathcal{C}}(s, X_s) = N_{\mathcal{C}}(s, X_s) - N_{\mathcal{C}}(s, 1 - X_s)$

- $(N_\alpha, \dots, N_\kappa)$ is a sufficient statistic for $(\alpha, \dots, \kappa)$

Baron, Lakshminarayan, Chen (2001)

Parameter estimation

The *maximum pseudo-likelihood estimator* of $\Theta = (\alpha, \lambda, \mu, \rho, \pi, \eta, \zeta, \theta, \iota, \kappa)$ is

$$\begin{aligned}\hat{\Theta} &= \arg \max_{\Theta} \prod_{s \in \mathcal{W}_0} P\{X_s \mid \mathcal{Q}_s\} \\ &= \arg \min_{\Theta} \prod_{s \in \mathcal{W}_0} \left(1 + \exp \left\{ \sum_{\text{clique types } c} c \Delta_c(s, X_s) \right\} \right)\end{aligned}$$

where X_s are conditionally independent on $\mathcal{W}_0 \subset \mathcal{W}$.

The *generalized maximum pseudo-likelihood estimator* is

$$\hat{\Theta} = \arg \max_{\Theta} \prod_{s \in \mathcal{W}} P\{X_s \mid \mathcal{Q}_s\}$$

with $\mathcal{W}_0 = \mathcal{W}$. It is consistent!

Comets (1992)

Application in semiconductor industry

$\hat{\alpha}$ means a “pure” failure rate

$$\frac{e^{\hat{\alpha}}}{1 + e^{\hat{\alpha}}} = P \left\{ \begin{array}{l} \text{a chip fails regardless} \\ \text{of the spatial interaction} \end{array} \right\}$$

$(\hat{\lambda}, \hat{\mu}, \hat{\rho}, \hat{\pi})$ determine “typical” shapes, sizes, directions of clusters

High values $\Rightarrow$ large clusters

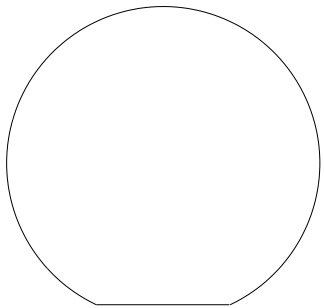
If $\hat{\lambda} \approx \hat{\mu}$ and $\hat{\rho} \approx \hat{\pi} \Rightarrow$ round clusters

If $\hat{\lambda} > \hat{\mu} \Rightarrow$ horizontally stretched clusters

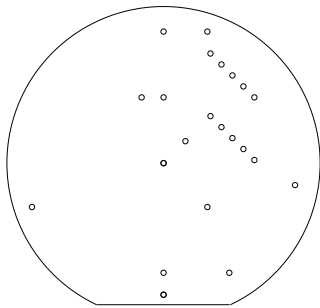
$\left(u = \max\{\hat{\lambda}/\hat{\mu}, \hat{\mu}/\hat{\lambda}\}, v = \max\{\hat{\rho}/\hat{\pi}, \hat{\pi}/\hat{\rho}\} \right) \sim$ “typical” direction

$$\text{Direction} \hat{=} \frac{e^u \mathbf{z}_1}{1 + e^u} + \frac{e^v \mathbf{z}_2}{1 + e^v}, \text{ where } \begin{array}{l} \mathbf{z}_1 \parallel \vec{u} \\ \mathbf{z}_2 \parallel \vec{v} \end{array}$$

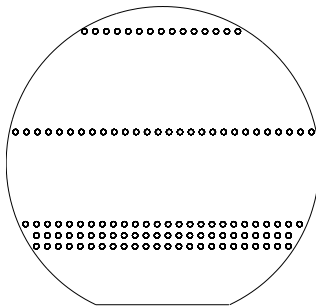
Examples



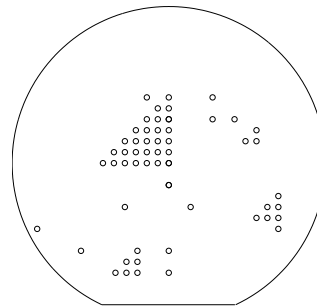
$$\alpha = -\infty$$



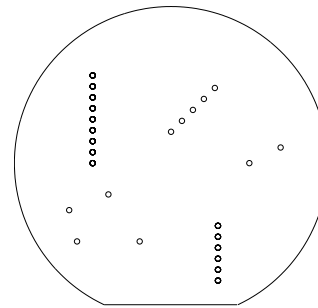
$$\rho > -\infty$$



$$\lambda > 0$$

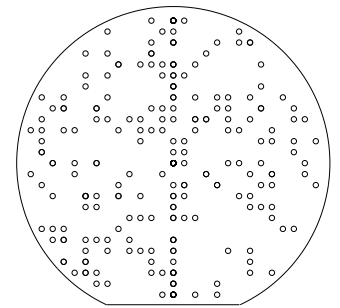


$$\eta > 0$$



$$\mu > -\infty$$

$$\pi > -\infty$$



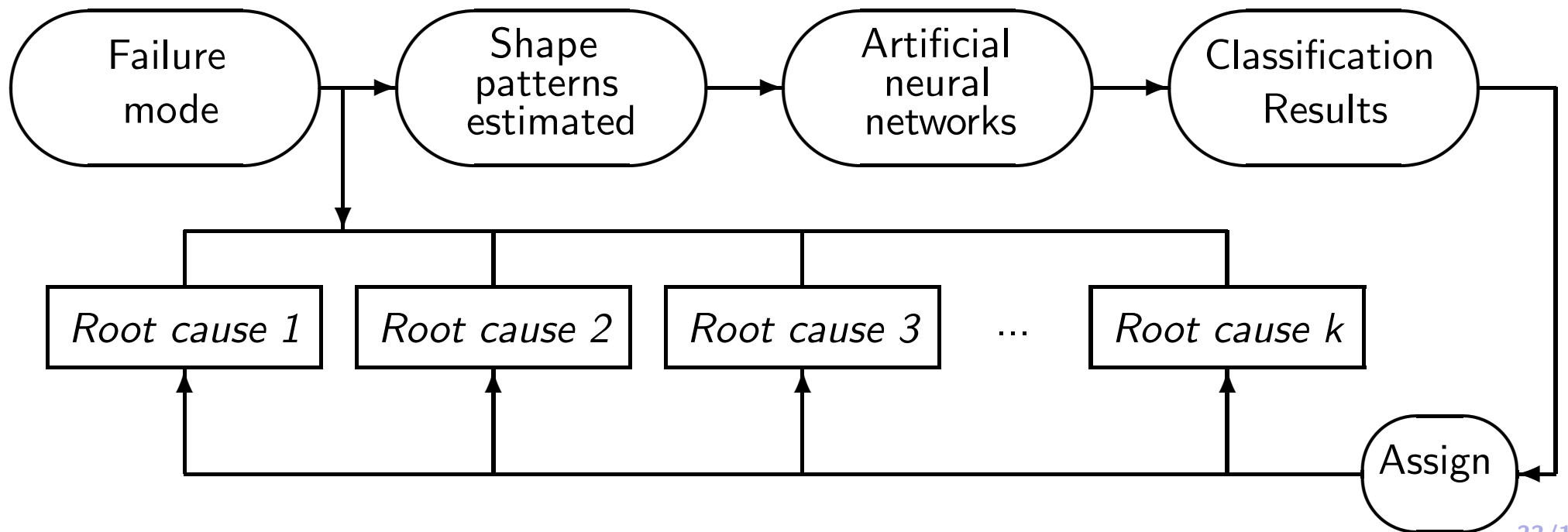
$$\left. \begin{matrix} \lambda, \mu \\ \rho, \pi \end{matrix} \right\} > -\infty$$

Matching patterns to root causes

Training data: 388 wafers representing all the known root causes.
Causes are determined by a direct in-depth engineering analysis.

The F test shows significant differences ($P = 0.0001$) between parameter estimates in different groups (root causes).

Matching algorithm



Application to 911 calls

$\hat{\alpha}$ means a “pure” rate of events

$$\frac{e^{\hat{\alpha}}}{1 + e^{\hat{\alpha}}} = P \left\{ \begin{array}{l} \text{a 911 call regardless} \\ \text{of the spatial interaction} \end{array} \right\}$$

$(\hat{\lambda}, \hat{\mu}, \hat{\rho}, \hat{\pi})$ determine direction and intensity of spread

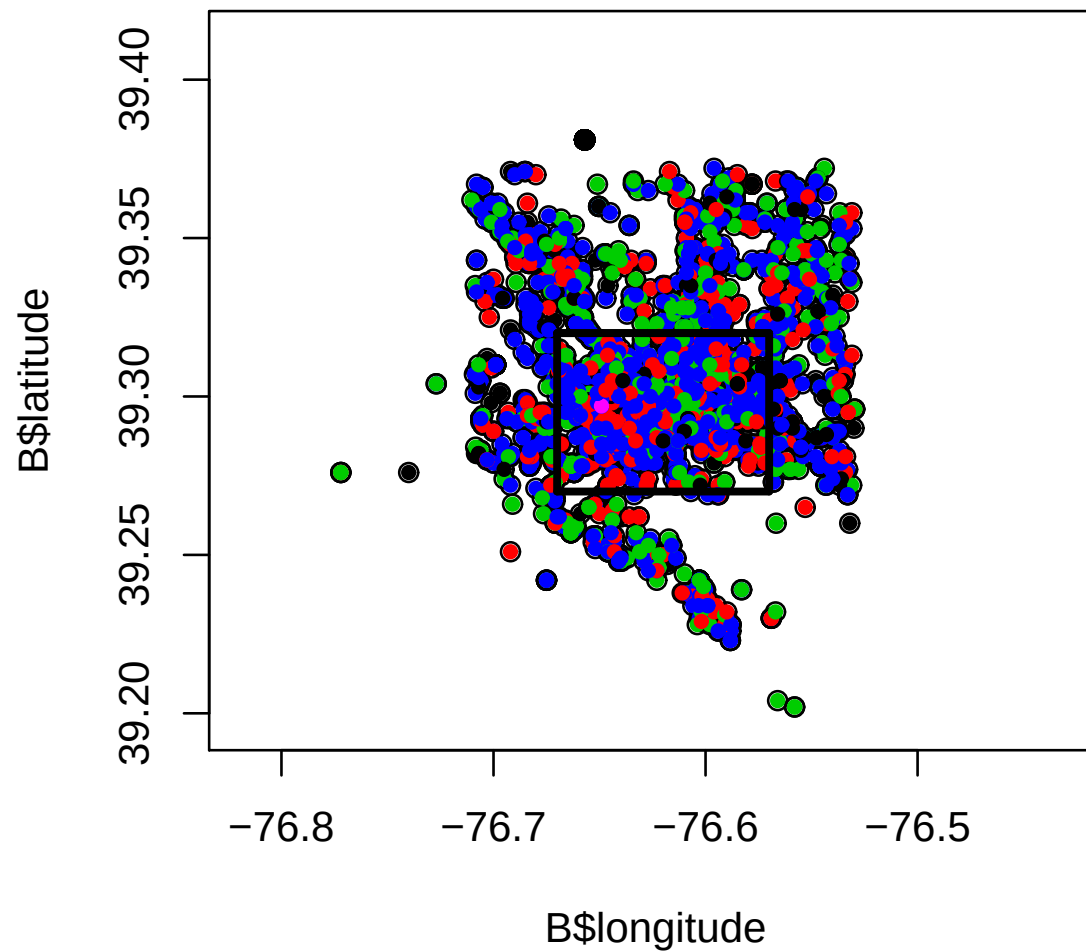
High values $\Rightarrow$ fast spread, strong interaction, possible organization

$\left(u = \max\{\hat{\lambda}/\hat{\mu}, \hat{\mu}/\hat{\lambda}\}, v = \max\{\hat{\rho}/\hat{\pi}, \hat{\pi}/\hat{\rho}\} \right) \sim \text{“typical” direction}$

$$\text{Direction} \hat{=} \frac{e^u z_1}{1 + e^u} + \frac{e^v z_2}{1 + e^v}, \text{ where } \begin{array}{l} z_1 \parallel \vec{u} \\ z_2 \parallel \vec{v} \end{array}$$

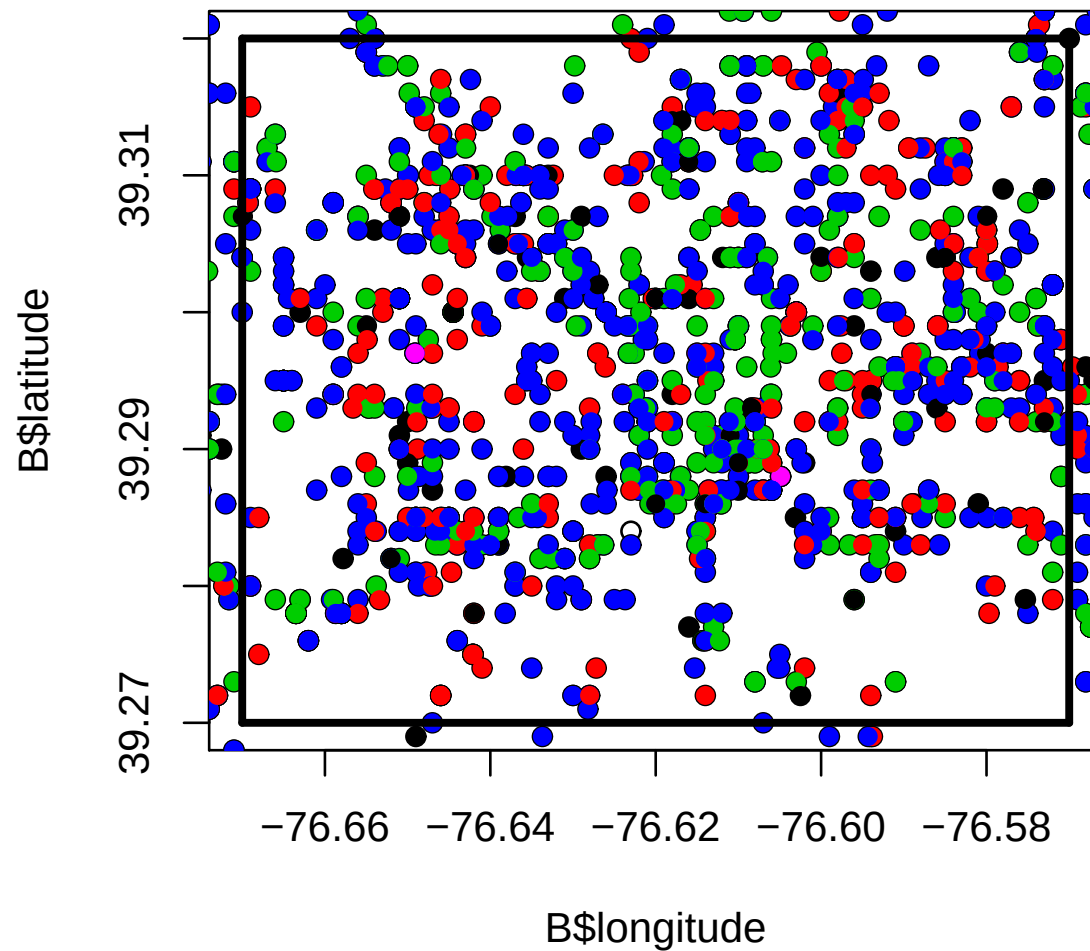
If $\hat{\lambda} \approx \hat{\mu}$ and $\hat{\rho} \approx \hat{\pi} \Rightarrow$ no clear direction

911 calls in Baltimore, 08/18/2017

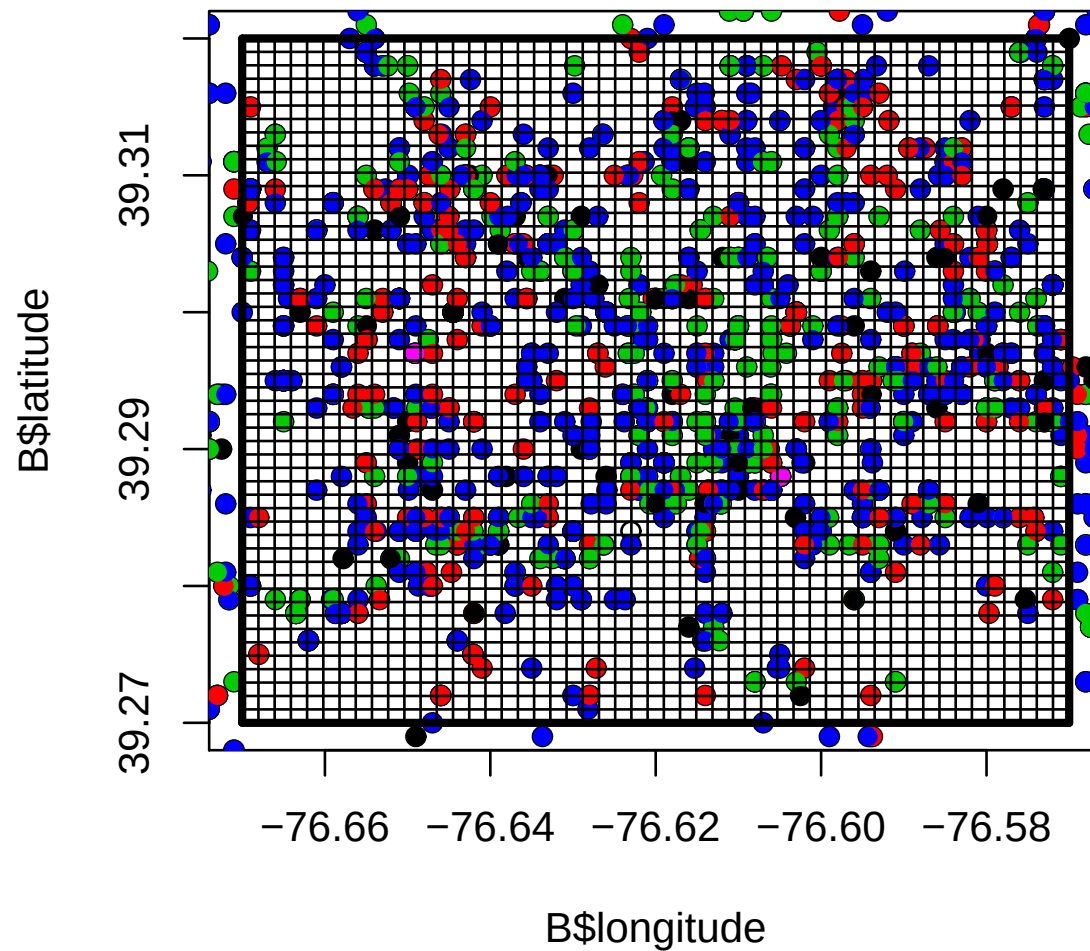


Priority	Non-emergency	Low	Medium	High	Emergency	Total
N	329	614	1382	515	2	2842

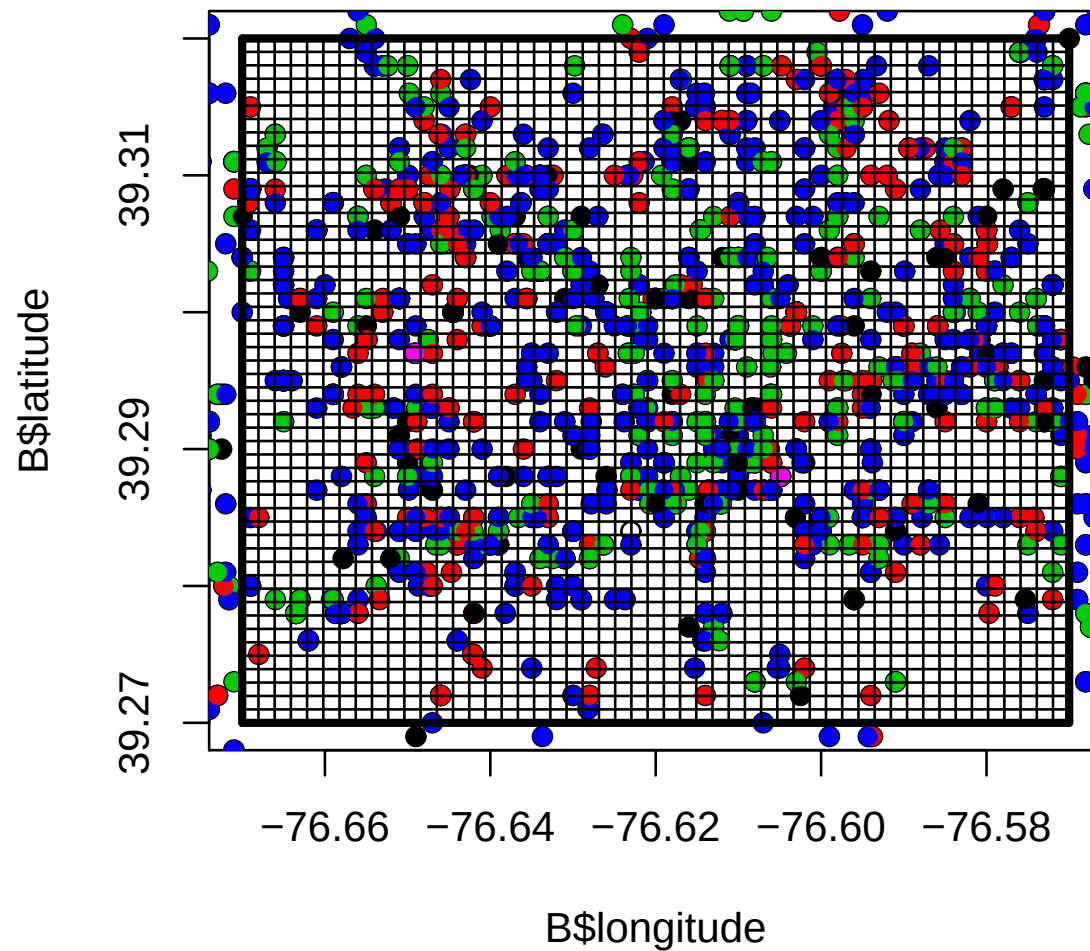
911 calls in Baltimore, 08/18/2017



911 calls in Baltimore, 08/18/2017

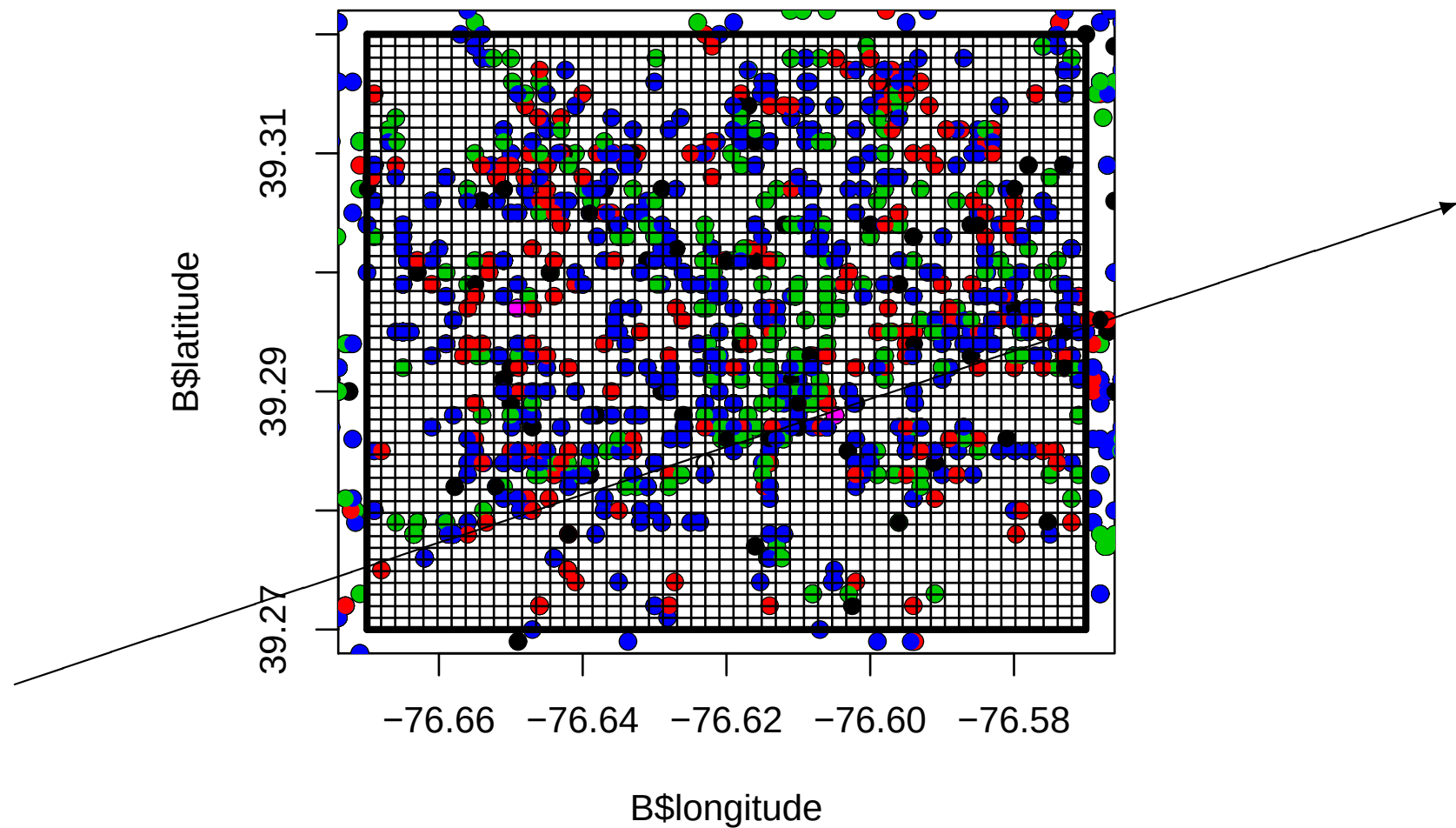


911 calls in Baltimore, 08/18/2017



$\hat{\alpha}$	$\hat{\lambda}$	$\hat{\mu}$	$\hat{\rho}$	$\hat{\pi}$	$\hat{\eta}$	$\hat{\theta}$	$\hat{\zeta}$	$\hat{\iota}$	$\hat{\kappa}$
-0.7	-1.3	-1.6	-1.3	-0.9	-1.7	-2.0	-2.5	-2.7	-3.2

911 calls in Baltimore, 08/18/2017



$\hat{\alpha}$	$\hat{\lambda}$	$\hat{\mu}$	$\hat{\rho}$	$\hat{\pi}$	$\hat{\eta}$	$\hat{\theta}$	$\hat{\zeta}$	$\hat{i}$	$\hat{\kappa}$
-0.7	-1.3	-1.6	-1.3	-0.9	-1.7	-2.0	-2.5	-2.7	-3.2

Change-Point Detection

Collect $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \dots)$ sequentially, where a random vector

$$\mathbf{X}_t = (X_{t1}, \dots, X_{td}) \in \mathcal{W}$$

is observed at time t .

A change in distribution occurs at time ν if

$$P\{\mathbf{X}_t \in A\} = \begin{cases} F(A) & \text{for } t \leq \nu \\ G(A) & \text{for } t > \nu \end{cases}, \quad \text{for some } A \subset \mathcal{W}$$

In semiconductor manufacturing, a change-point means deviation from pre-set parameters, appearance of a new root cause.

In a sequence of social events, a change-point may indicate a new trend, change of direction, global unrest, organized violence.

Change-Point Detection

Looking for T , a *stopping time*, which

- ▶ minimizes the *mean delay*

$$\text{MD} = \mathbf{E}_{\nu}(T - \nu)^+$$

- ▶ maximizes the *mean time between false alarms*

$$\text{MTBFA} = \mathbf{E}\{T | \text{no change}\} = \mathbf{E}_{\nu=\infty} T$$

Change-Point Detection

Classical sequential scheme - CUSUM

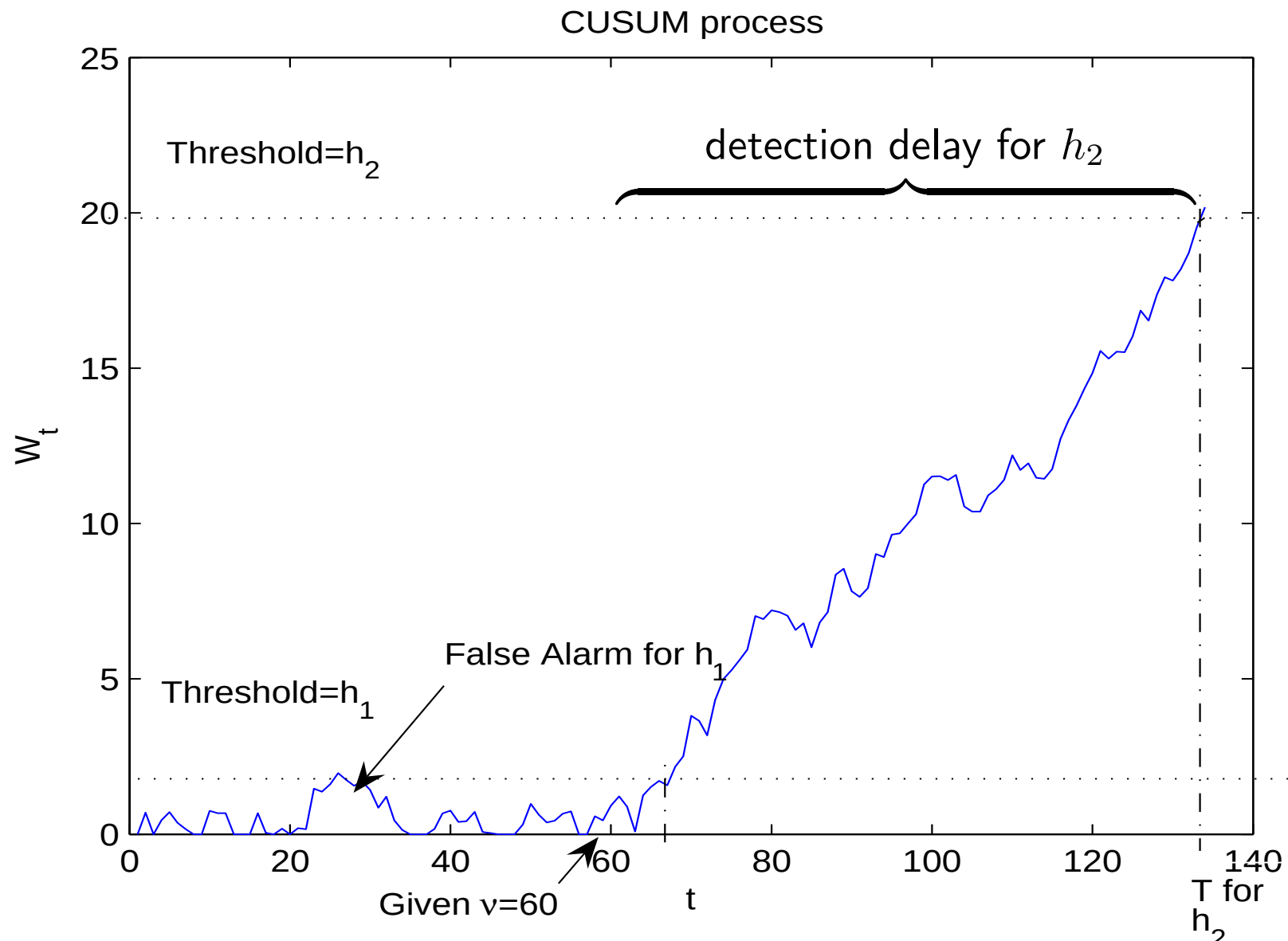
When pre- and after-change distributions F , G are known, and $\{X_n\}$ are i.i.d., **CUSUM stopping rule** is optimal in various ways.

$$W_n = \max_t \sum_{j=t+1}^n \log \frac{g(X_j)}{f(X_j)}, \quad T_W(h) = \inf \{n : W_n \geq h\}$$

For non-i.i.d. data, $W_n = \max_t \sum_{j=t+1}^n \log \frac{g(X_j|X_1, \dots, X_{j-1})}{f(X_j|X_1, \dots, X_{j-1})}$

Page 1954, Lorden 1971, Moustakides 1986, Ritov 1990

Change-Point Detection



Change-Point Detection - Pseudolikelihood Approach

When the joint density f is not available, we may substitute it by the *pseudolikelihood function*

$$\tilde{f}(\mathbf{X}) = \prod_{s \in \mathcal{W}} P \{X_s \mid \mathcal{Q}_s\}$$

Pseudo-CUSUM process

$$\begin{aligned} \tilde{W}_n &= \max_t \sum_{i=t}^n \log \frac{\tilde{g}(\mathbf{X}_i)}{\tilde{f}(\mathbf{X}_i)} = \max_t \sum_{i=t}^n \sum_{s \in \mathcal{W}} \log \frac{P_G \{X_{is} \mid \mathcal{Q}_{is}\}}{P_F \{X_{is} \mid \mathcal{Q}_{is}\}} \\ &\leq \sum_{s \in \mathcal{W}} \max_{t_s} \sum_{i=t_s}^n \log \frac{P_G \{X_{is} \mid \mathcal{Q}_{is}\}}{P_F \{X_{is} \mid \mathcal{Q}_{is}\}} = V_n \end{aligned}$$

Change-Point Detection - Sum of CUSUM

V_n is the sequential test statistic of

$$H_{0,n} : \min_s \nu_s > n \quad \text{vs} \quad H_{A,n} : \min_s \nu_s \leq n$$

CUSUM stopping rule

$$T_W = \min\{n : H_{0,n} \text{ is rejected}\} = \min\{n : W_n \geq h\}, \text{ given } \nu_s \equiv \nu.$$

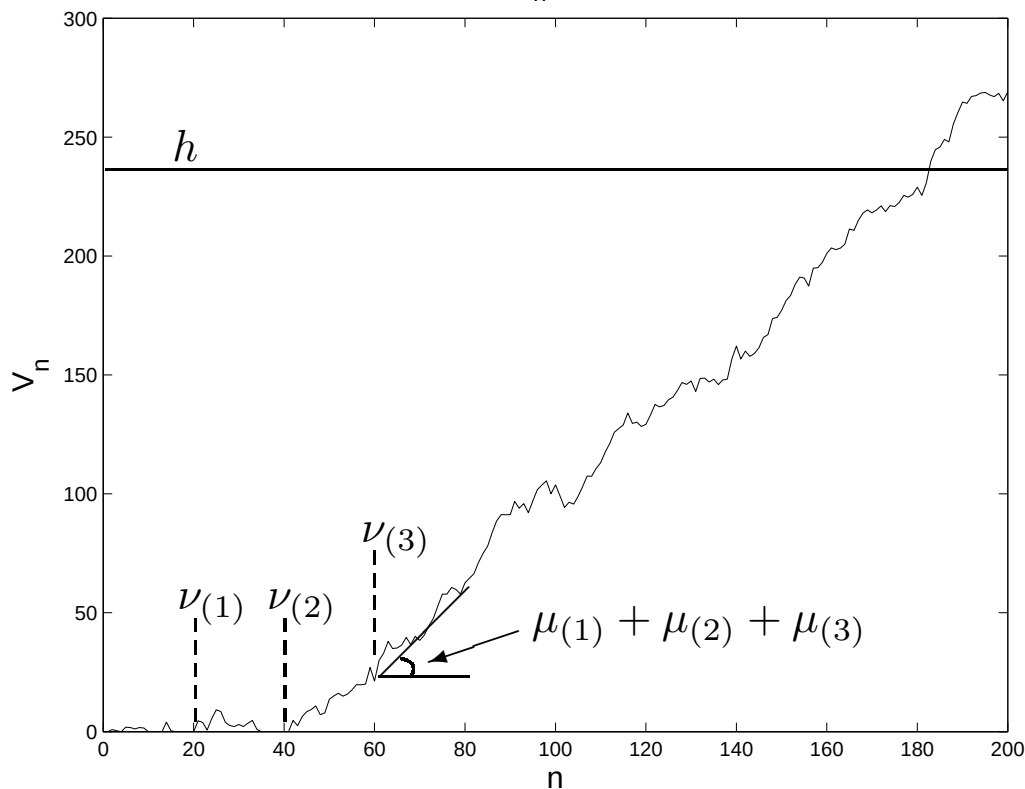
Sum of CUSUM

$$T_V = \min\{n : H_{0,n} \text{ is rejected}\} = \min\{n : V_n \geq h\}, \text{ general case}$$

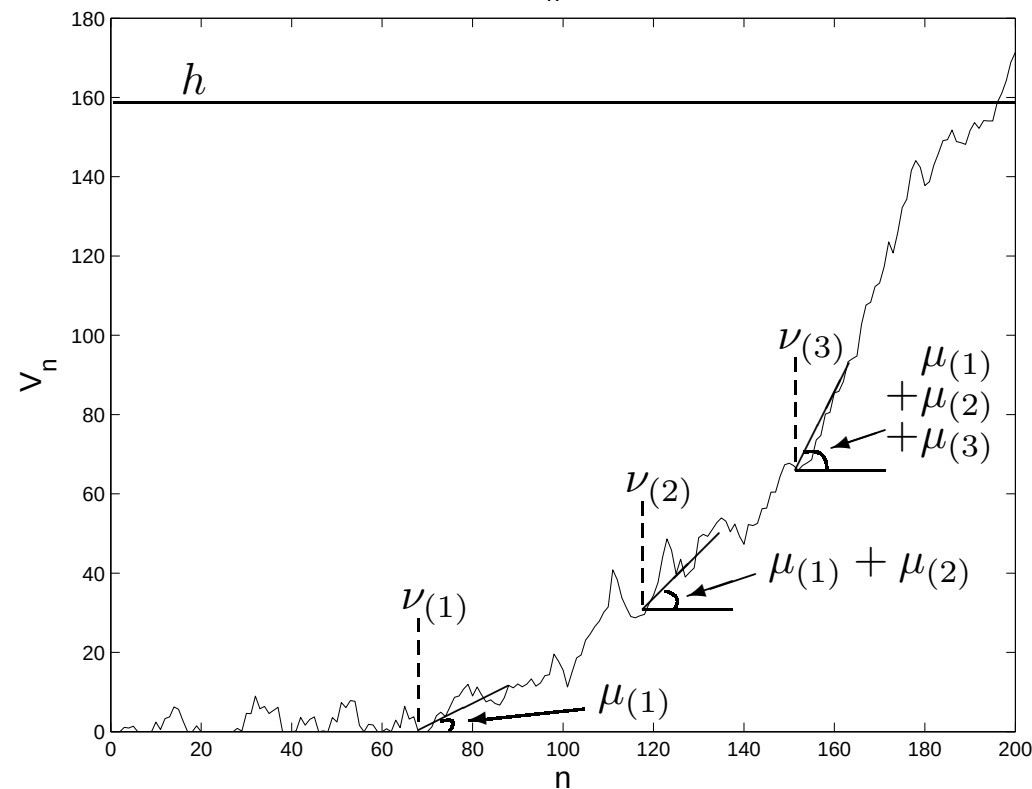
$$V_n = \sum_s W_n^{(s)}, \text{ sum of marginal CUSUM}$$

Change-Point Detection - Sum of CUSUM

Sum-CUSUM process V_n with "early" change-points



Sum-CUSUM process V_n with "rare" change-points



Delay of Sum-CUSUM - “Early” Change-Points

Suppose that ℓ change-points occur at time intervals of the lower order than threshold h , as $h \rightarrow \infty$,

$$\nu_{(1)} = o(h), \nu_{(2)} - \nu_{(1)} = o(h), \dots, \nu_{(\ell)} - \nu_{(\ell-1)} = o(h).$$

Then, for the stopping time $T_V(h) = \inf \{n : V(h) \geq h\}$,

$$\frac{\text{Delay}(h)}{h} = \frac{(T_V(h) - \nu_{(1)})^+}{h} \rightarrow \frac{1}{\mu_{(1)} + \dots + \mu_{(\ell)}},$$

almost surely, as $h \rightarrow \infty$, where

$$\mu_{(j)} = K(G_j, F_j) = E_G \log \{f_j(X)/g_j(X)\} > 0$$

Delay of Sum-CUSUM - “Rare” Change-Points

Suppose that change-points occur at time intervals of the order comparable with threshold h , $\nu_{(k)} = q_k h + o(h)$ for $q_1 < q_2 < \dots$

Then, for the Sum-CUSUM process V_n with stopping time $T_V(h)$

$$\frac{\text{Delay}(h)}{h} = \frac{(T_V(h) - \nu_{(1)})^+}{h} \rightarrow \frac{1 + \sum_{k=1}^m (q_k - q_1) \mu_{(k)}}{\sum_{k=1}^m \mu_{(k)}}$$

almost surely, as $h \rightarrow \infty$, where

$$m = \max \left\{ k : \sum_{j=1}^k (q_k - q_j) \mu_{(j)} \leq 1 \right\}$$

Change-Point Detection - Moments of the Delay

“Early” change-points $\implies$

$$\limsup_{h \uparrow \infty} \frac{E(\text{Delay})^p}{h^p} = \limsup_{h \uparrow \infty} \frac{E \{ (T_V(h) - \nu_{(1)})^+ \}^p}{h^p} \leq \frac{1}{(\mu_{(1)} + \dots + \mu_{(\ell)})^p}$$

“Rare” change-points $\implies$

$$\limsup_{h \uparrow \infty} \frac{E(\text{Delay})^p}{h^p} = \limsup_{h \uparrow \infty} \frac{E \{ (T_V(h) - \nu_{(1)})^+ \}^p}{h^p} \leq \left(\frac{1 + \sum_{k=1}^m (q_k - q_1) \mu_{(k)}}{\sum_{k=1}^m \mu_{(k)}} \right)^p$$

for all $p > 0$, where $m = \max \left\{ k : \sum_{j=1}^k (q_k - q_j) \mu_{(j)} \leq 1 \right\}$.

Use: choose threshold h based on the desired mean delay.

Change-Point Detection - Rate of False Alarms

$$\text{RFA} = (\text{Mean time between false alarms})^{-1} = 1/\mathbf{E}_{\nu=\infty}(T)$$

$$\text{RFA} = \mathbf{E} \{T_V(h) \mid \nu_j = \infty \text{ for } j = 1, \dots, |\mathcal{W}|\} \geq e^{h/|\mathcal{W}|}$$

and asymptotically, as $h \rightarrow \infty$,

$$\text{RFA} \sim \sum_{s \in \mathcal{W}} \sum_{\{s_1, \dots, s_r\} \subset \mathcal{W}} (\pi_{s_1}(h) * \dots * \pi_{s_r}(h)),$$

$$\pi_s(h) \sim \frac{P_G \{z_s > 0\} - P_F \{z_s > 0\}}{P_F \{z_s \geq h\}} + e^{-h}$$

$$z_s = \log \frac{\mathbf{P}_G \{X_s \mid \mathcal{Q}_s\}}{\mathbf{P}_F \{X_s \mid \mathcal{Q}_s\}} = \log \frac{1 + \exp \{-\alpha_G \Delta_\alpha(s) - \dots - \kappa_G \Delta_\kappa(s)\}}{1 + \exp \{-\alpha_F \Delta_\alpha(s) - \dots - \kappa_F \Delta_\kappa(s)\}},$$

from the Markov random field

Comparison

Mean delays with threshold h chosen to have $\text{RFA} \approx 0.005$.

Pseudo-CUSUM ($h = 9.88$)				Sum-CUSUM ($h = 6.26$)		
Changing parameters						
$ \theta_1 - \theta_0 $	λ, μ, ν	μ, ν	λ	λ, μ, ν	μ, ν	λ
0.00	200.35	200.35	200.35	200.17	200.16	200.17
0.25	38.28	64.80	111.45	31.09	44.95	76.11
0.50	11.55	24.81	59.63	11.67	16.75	30.26
0.75	5.17	11.62	39.36	6.26	8.57	16.19
1.00	3.22	6.54	23.57	4.31	5.73	10.81
1.25	2.34	4.34	15.93	3.28	4.44	7.55
1.50	1.93	3.22	11.39	2.59	3.52	5.95
1.75	1.57	2.58	8.46	2.22	3.03	5.12
2.00	1.35	2.17	6.28	1.97	2.59	4.32

References

- C. A. J. Acosta-Mejia. *Pseudolikelihood Methods in Multichannel Change-Point Detection*, Ph. D. dissertation, The University of Texas at Dallas, 2016.
- M. Baron, C. K. Lakshminarayan, and Z. Chen. Markov random fields in pattern recognition for semiconductor manufacturing. *Technometrics*, 43(1):66–72, 2001.
- J. Besag. Spatial interaction and the statistical analysis of lattice systems. *J. Royal Statist. Soc.*, 2:192–236, 1974.
- F. Comets. On consistency of a class of estimators for exponential families of Markov random fields on the lattice. *Ann. Stat.*, 20:455–468, 1992.
- ▶ G. Lorden. Procedures for reacting to a change in distribution. *Ann. Math. Statist.*, 42:1897–1908, 1971.
- ▶ E. S. Page. Continuous inspection schemes. *Biometrika*, 41:100–115, 1954
- ▶ H. V. Poor and O. Hadjiliadis. *Quickest Detection*. Cambridge University Press, Cambridge, UK, 2009
- ▶ Y. Ritov. Decision theoretic optimality of CUSUM procedure. *Ann. Stat.*, 18:1464–1469, 1990.
- ▶ A. G. Tartakovsky, I. V. Nikiforov, and M. Basseville. *Sequential Analysis: Hypothesis Testing and Change-Point Detection*. Chapman & Hall/CRC, 2014

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Thank you!

